Calendar

Combinatorics

Fall 2026

I will put all deadlines on the Blackboard Calendar. Please check it regularly.

Key university dates

For official University dates, see the Fall 2026 academic calendar and the Registrar's final-exam schedule.

Date	Event
Monday, August 17	Classes begin
Monday, September 7	Labor Day holiday
Tuesday, November 24–Friday, November 27	Thanksgiving holiday
Wednesday, December 9	Final exam: Wednesday, December 9, 2026, 8:00-10:00

Exams

Date	Assessment
Friday, September 18	Midterm 1
Friday, October 16 TENTATIVE	Midterm 2
Friday, November 13 TENTATIVE	Midterm 3
Wednesday, December 9, 2026, 8:00-10:00	Exam 4 and mastery reassessment

Daily lecture log

The public daily log below contains the lectures currently released. Later planning entries remain in the instructor's private lecture log and do not appear in this navigation.

Wk	Lec	Topics	Ref
1	**L1. Set notation and basic counting** 2026-08-17	Sets, cardinality, Cartesian products, and the sum rule.	Set notation and basic counting: from "Set notation" through "The sum rule."
	L2. Product rule and strings 2026-08-19	Product rule; strings with and without repetition.	Set notation and basic counting: sections "The product rule" and "Strings: an important type of counting problems."

Wk	Lec	Topics	Ref
	L3. More counting techniques 2026-08-21	Subtraction, division, permutations, combinations, and counting models.	More counting techniques: sections "The difference rule," "The division principle and permutations," "Unordered selections: combinations without repetition," and "Counting problems."
2	**L4. Applications of combinations and combinations with repetition** 2026-08-24	Lattice paths. Repeated-letter anagrams and multinomial coefficients. Combinations with repetition; stars and bars.	Applications of combinations and combinations with repetition: sections "Lattice paths in the plane," "Three-dimensional paths and multinomial coefficients," "Anagrams with repeated letters," and "Combinations with repetition."
	L5. Review of choice models, increasing sequences, and basic binomial expansions 2026-08-26	Choosing the correct model: does order matter and are repetitions allowed. A second bijection for combinations with repetition: increasing integers. Integer solutions with lower bounds. The binomial formulas for $(x+y)^n$. Sum of all $C(n,k) = 2^n$.	Binomial expansions, Pascal paths, and Catalan numbers: sections "A bakery problem: four kinds of choice," "Combinations with repetition and cumulative totals," "Solutions with lower bounds," and "The binomial theorem."
	L6. Polynomials, paths, Pascal recursion and restricted lattice paths, intro to inclusion–exclusion 2026-08-28	Multinomial formula and polynomial expansions Coefficient extraction from powers of multivariate polynomials. Pascal's recurrence and triangle: algebraic and combinatorial interpretation. Recursive path counts; paths through or avoiding a specified point; recursive definition; subtraction arguments.	Binomial expansions, Pascal paths, and Catalan numbers: sections "The multinomial theorem," "Pascal's recurrence and Pascal's triangle," and "Recursive lattice-path counts."
3	**L7. Catalan paths and the inclusion–exclusion principle** 2026-08-31	Reflection argument and Catalan numbers. Inclusion–exclusion for two, three, and finitely many sets.	Binomial expansions, Pascal paths, and Catalan numbers: section "Catalan paths and the reflection principle." Inclusion–exclusion and applications: from "The geometry of overlapping sets" through "Proof via indicator functions."
	L8. Applications of inclusion–exclusion 2026-09-02	Characteristic functions. Conclusion of the proof of the inclusion–exclusion principle. Complement and violation-set formulations.	Inclusion–exclusion and applications: sections "Proof via indicator functions" and "The complement and violation-set form."
	L9. Applications of inclusion–exclusion 2026-09-04	Applications to bounded integer solutions and strings without repeated letters that avoid or contain specified substrings. Euler's totient function.	Inclusion–exclusion and applications: sections "Integer solutions with upper bounds," "Strings with forbidden substrings," and "Euler's totient function."

Wk	Lec	Topics	Ref
4	**L10. Applications of inclusion–exclusion** 2026-09-09	Review of counting strings that avoid specified substrings. The inclusion–exclusion formula when an intersection size depends only on the number of conditions imposed. Derangements.	Inclusion–exclusion and applications: sections "Strings with forbidden substrings," "When only the number of imposed conditions matters," and "Derangements and fixed points."
	L11. Applications of inclusion–exclusion 2026-09-11	Strings that use every symbol, surjective functions, and Stirling numbers. Graphs without isolated vertices.	Inclusion–exclusion and applications: sections "Euler's totient function," "Graphs with no isolated vertices," "Counting surjective functions and full-alphabet strings," and "Connection with Stirling numbers."
5	**L12. Rook polynomials and forbidden positions** 2026-09-14	Review of rook numbers and rook polynomials, including examples of their computation and use. Introduction to ordinary generating functions: a generating function $\sum_{n\geq 0} a_n x^n$ records the count a_n at value n. The Cauchy product of formal power series, including a formal justification for exchanging the order of summation. Counting ways to make 10 cents using 1-, 2-, and 5-cent coins. We first used separate variables to record the number of each type of coin, then substituted t, t^2, and t^5 so that the exponent records total value. We also discussed truncating the generating functions when only a fixed target value is needed.	Rook polynomials and forbidden positions: all sections. Ordinary generating functions: definitions and series multiplication: all sections.
	Midterm 1 2026-09-18	Midterm 1.	
6	**L13. Ordinary generating functions: definitions and series multiplication** 2026-09-21	Ordinary generating functions; encoding finite and unlimited selections; substitution to track total value; formal power-series multiplication and the Cauchy product; generating-function models for selections with conditions.	Ordinary generating functions: definitions and series multiplication: all sections.
	L14. Ordinary generating functions: advanced techniques 2026-09-23	Infinite formal power series and rational functions; constrained integer equations; combinatorial interpretations of powers and products; operations on generating functions; application to the sum of squares.	Ordinary generating functions: advanced techniques: all sections except coefficient extraction.

Wk	Lec	Topics	Ref
	TODO: Friday 9/25 2026-09-25 TENTATIVE		
7	**TODO: Monday 9/28** 2026-09-28 TENTATIVE		
	TODO: Wednesday 9/30 2026-09-30 TENTATIVE		
	L18. Solving homogeneous linear recurrences with constant coefficients 2026-10-02	Introduction to solving homogeneous linear recurrences with constant coefficients: classification of recurrences and the first steps toward closed formulas.	Solving linear recurrences with constant coefficients: sections "Review: classifying recurrence relations" and "Homogeneous recurrences with real characteristic roots" (subsection "Distinct real roots and the opening problem").
8	**No class** 2026-10-05	No class.	
	L19. Characteristic polynomials, repeated roots, and nonhomogeneous recurrences 2026-10-07	The characteristic polynomial of a linear recurrence with constant coefficients. Solutions of homogeneous recurrences; roots with multiplicity. Nonhomogeneous recurrences: particular solutions and the general solution.	Solving linear recurrences with constant coefficients: sections "Homogeneous recurrences with real characteristic roots" (subsections "Repeated real roots and the shift operator" and "Summary of the homogeneous solution procedure") and "Nonhomogeneous polynomial–exponential driving terms."
	TODO: Friday 10/9 2026-10-09 TENTATIVE		
9	**TODO: Monday 10/12** 2026-10-12 TENTATIVE		
	L22. Finite state machines 2: recurrence relations and fast computation 2026-10-14	From a transition-matrix (vector) recurrence to a scalar recurrence relation. Computing terms of a recurrence with a logarithmic number of matrix multiplications.	Finite state machines and vector recurrences: sections "From a vector recurrence to a scalar recurrence" and "Computing matrix powers."

Wk	Lec	Topics	Ref
	Midterm 2 2026-10-16	Midterm 2.	
10	**Fall break** 2026-10-19	No class.	
	L23. Groups, symmetries of polygons, and permutations 2026-10-21	Colorings of the corners of a square up to rotation, and the definition of a group. Dihedral groups: the symmetries of a regular polygon, the rotation r and the reflection s. Presentations by generators and relations. The symmetric group S_n; cycles and cycle notation.	Groups: sections "Coloring the corners of a square," "Groups," "The symmetries of a regular polygon," "Presentations," "The symmetric group," and "Cycle notation."

TENTATIVE Lecture topics and assessment dates may change.

Week and unit overview

Weeks	Unit	Topic
1–4	Unit I: Enumeration	Counting models, bijections, double counting, and inclusion-exclusion
4–5	Unit II: Generating-function foundations	Formal power series and ordinary generating-function operations
6–8 TENTATIVE	Unit III: Recurrences	State models, linear recurrences, recursive evaluation, and dynamic programming
9–16 TENTATIVE	Further topics	To be announced